

Computer Networks & Security

UNIT-01

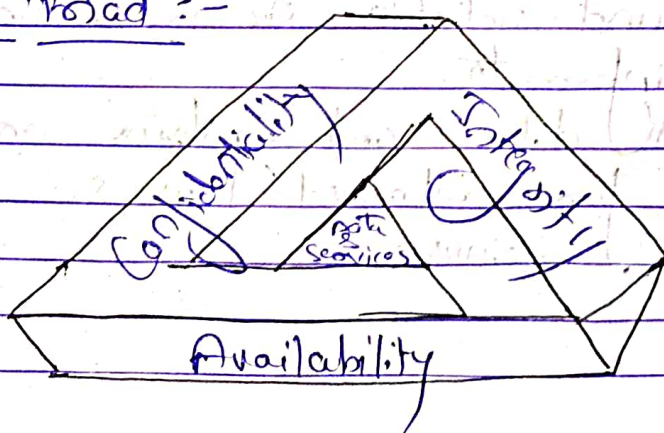
CLASSICAL ENCRYPTION TECHNIQUES

- Q1. Define of Computer Security?
Q2. Explain CIA triad?

Ans) Computer Security :-

The protects afforded to an automated information system in order to attain the applicable objective of preserving the integrity, availability and confidentiality of information system resources.
(includes hardware, software, firmware, information/data and telecommunication)

CIA Triad :-



Date _____
Page _____

* Confidentiality :-
(only legitimate party know)

- Ensure only authorized people can access sensitive information
- Deal with a lot of sensitive information
- Cannot afford to lose • Ex; Account information

* Integrity :-

- Maintaining the accuracy and trustworthiness of information
- Ensures the data remains unaltered
- Ensure that data remain intact
- Ex; Patient's information

* Availability :-

- Ensuring timely and reliable access to and use of information
- Making sure that data and systems are always available for authorized users
- Ex; Authentication service

→ Some additional concepts :-

* Authenticity - The property of being genuine and being able to be verified and trusted; confidence in validity of a transmission, a msg, or msg originator.

* Accountability - The security goal that generates the requirement for acts of an entity to be traced uniquely to that entity.
(Ensures users are not misusing the privileges)

→ Three levels of impact of security breach :-

* Low :- The loss could be expected to have a limited adverse effect on organizational operations, organizational assets, or individuals.

* Medium :- The loss could be expected to have a serious adverse effect on organizational operations, organizational assets, or individuals.

* High :- The loss could be expected to have a severe or catastrophic adverse effect on
" " " "

Cryptographic algorithms and protocols:-

① Symmetric encryption:-

- Uses a single key for both encryption and decryption.
- Used to conceal the contents of blocks or streams of data of any size, including msgs, files, encryption keys, & passwords.

② Asymmetric encryption:-

- Uses a pair of keys: a public and private key.
Public key; encryption
Private key; decryption
- Used to conceal small blocks of data, such as encryption keys and hash function values, which are used in digital signatures.

③ Data integrity algorithm:- Used to protect blocks of data, such as msgs from alteration.

④ Authentication protocols:- These are schemes based on the use of cryptographic algorithms designed to authenticate the identity of entities.

Solve $P(\text{Access built}) = P(B)$

The OSI Security Architecture

Threat $\rightarrow$ a threat is possible dangers that might exploit a vulnerability.

Attack $\rightarrow$ an intelligent act that is a deliberate attempt to evade (bachna) security services and violate the security policy of a system.

$\Rightarrow$ The OSI Security Architecture deals with the term :-

- ① Security Attack
- ② Security Mechanism
- ③ Security Services

Q3 What are possible security attack?

Q4 Explain Security services and security mechanisms

Q5 Differentiate passive attack from active attack with example.

Security Attack :-

• Action that compromises the security of an individual or an organization.

• Types

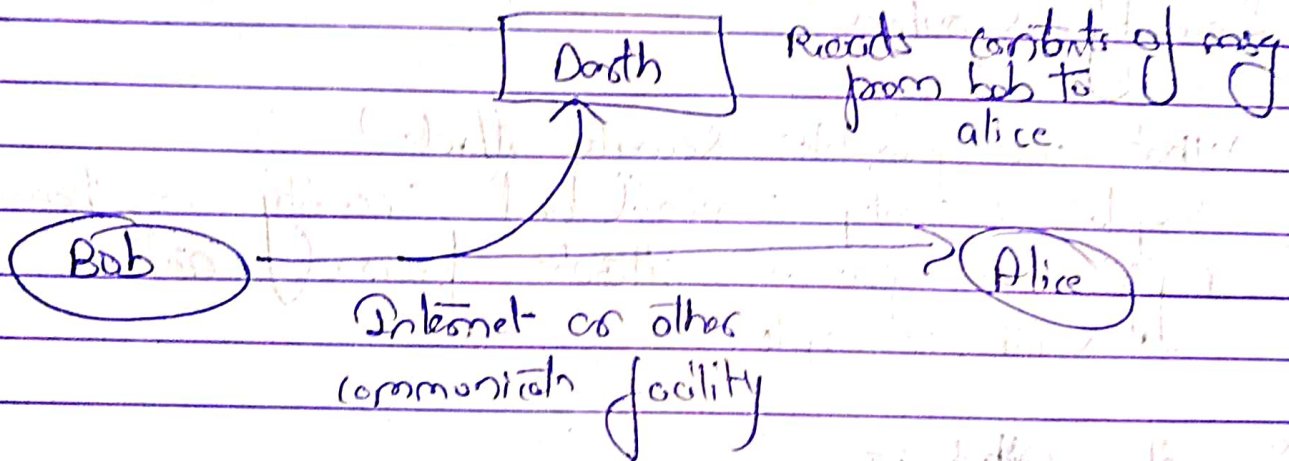
- (a) Passive attack
- (b) Active attack

① Passive Attack

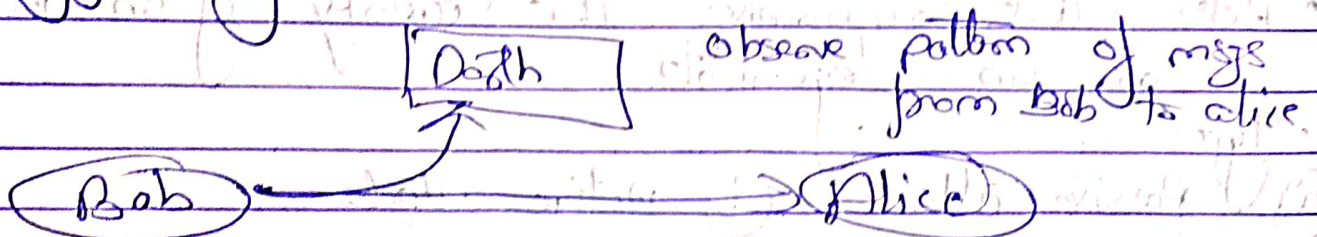
- Attempts to learn or make use of information from the system.
- Does not affect system resources.
- Eavesdropping or monitoring of transmissions.
- Goal: Obtain information that is being transmitted.

Types:-

① Release of msg contents



② Traffic analysis

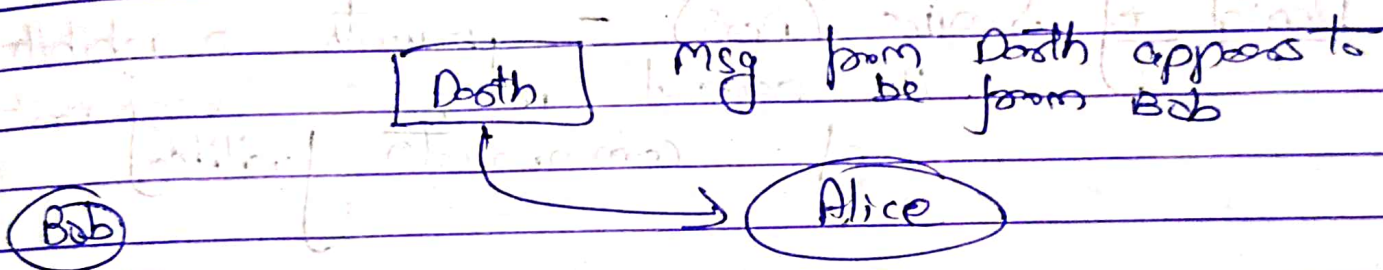


② Active Attack

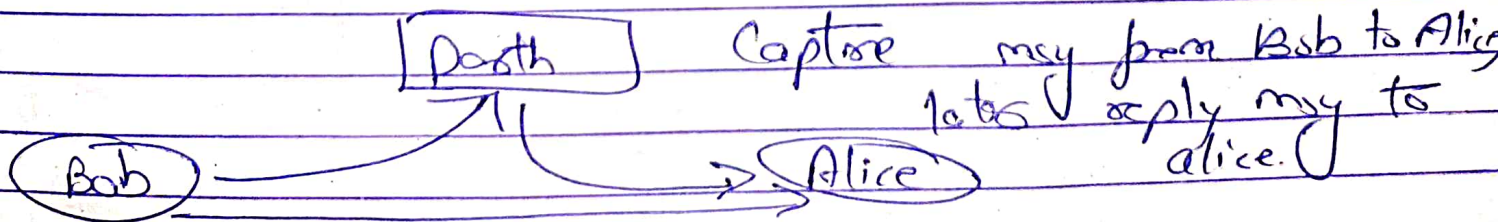
- Active attacks involve some modification of the data stream or the creation of a false stream.

Types:-

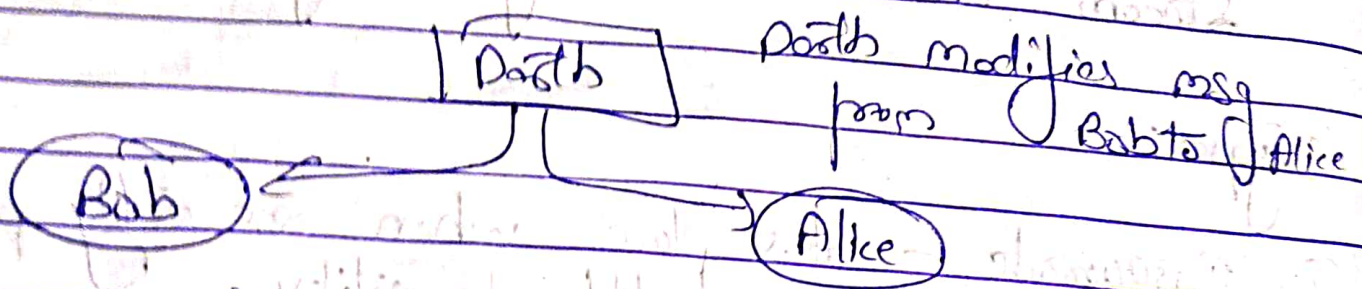
- ① Masquerade - takes place when one entity pretends to be a different entity.



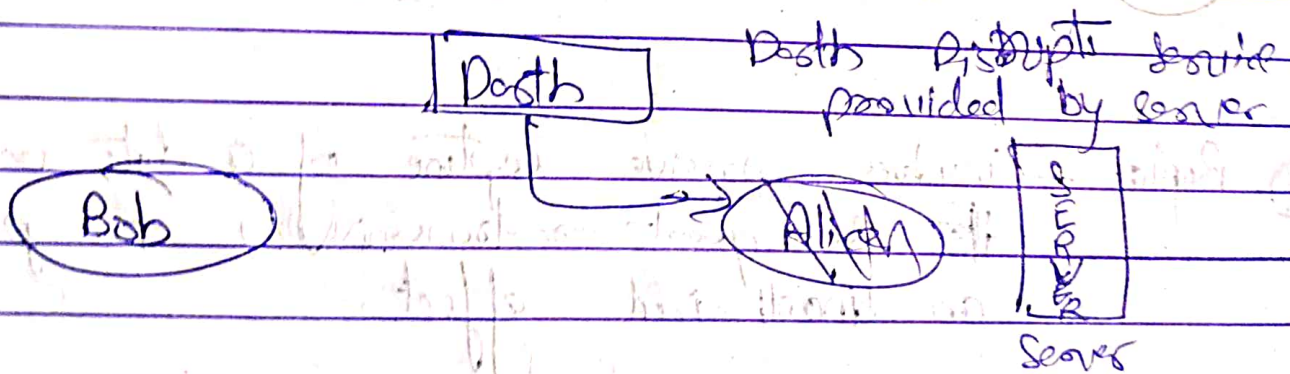
- ② Replay - involves passive capture of a data unit and its subsequent re-transmission to produce an unauthorized effect.



③ Modification of msg - Simply means that some parts of a legitimate msg is altered.



④ Denial of Service (DoS) - prevents or inhibits the normal use or management of communication facilities.



Solve A $P(\text{correct build}) = P(B)$
 $P(A|B) = P$

Passive Attack

- ① Hard to detect
- ② Neither sender nor receiver is aware of the attack
- ③ Encrypts prevents the success of the passive attacks
- ④ More emphasis is on prevention than detection
- ⑤ Types -
 - (a) Release of msg contents
 - (b) traffic analysis
- ⑥ Ex; Eavesdropping

Active Attack

- ① Hard to prevent
- ② Difficult to prevent - physical, software and methods vulnerabilities
- ③ Detect and recover from any disruption or delays
- ④ If the detector has a deterrent effect, it may also contribute to prevention
- ⑤ Types -
 - (a) modification of msg
 - (b) Replay
 - (c) Masquerade
 - (d) DoS attack
- ⑥ Ex; Man-in-the-middle Attack (MITM)

Security Services :-

The processing or communication service that is provided by a system to give a specific kind of protection to a system or resource. Security services implement security policies and are implemented by security mechanisms.

① Authentication - The assurance that the communicating entity is the one that it claims to be.

② Peer Entity Authentication - used in association with a logical connection to provide confidence in the identity of the entities connected.

③ Data Origin Authentication - In a connectionless transfer, provides assurance that the source of received data is claimed.

~~Solve~~ 1) $P(\text{concerns bail}) = P(B)$

$P(\text{convicted}) = P(C)$

$P(B|C) = \frac{P(B \cap C)}{P(C)}$

② Access Control - The prevention of unauthorized use of a resource

③ Data Confidentiality - The protection of data from unauthorized disclosure.

④ Connect Confidentiality - the protection of all user data on a connect

⑤ Connectless Confidentiality - the protection of all user data in a single data block

⑥ Selective-field Confidentiality - The confidentiality of selected fields within the user data on a connect or in a single data block.

⑦ Traffic-flow Confidentiality - the protection of the information that might be derived from observation of traffic flows.

④ Data Integrity - The assurance that data received are exactly as sent by an authorized entity.

① Connect Integrity with Recovery - provides for the integrity of all user data and a connect and detects any modifications, insert, delete, or replay of any data within an entire data sequence, with recovery attempted.

② Connect Integrity w/o recovery - same as connect integrity with recovery, but provides only detects w/o recovery.

③ Selective - Field Connect Integrity - refers to ensuring the integrity of specific within a data transmission, rather than the entire msg.

④ Connectless Integrity - provides for the integrity of a single connectless data block

~~Solve A $P(\text{concerns built}) = P(B)$~~

~~$P(\text{concerns}) = P$~~

~~$P(\text{concerns}) P(M)$~~

③ Nonrepudiation - provides protection against denial by one of the entities involved in a communication of having participated in all or part of the communication.

④ Nonrepudiation, origin - proof that the msg was sent by the specified party

⑤ Nonrepudiation, destination - proof that the msg was received by the specified party

Security Mechanisms :-

① Specific security mechanisms - may be incorporated into the appropriate protocol layers in order to provide some of the OSI security services.

② Encipherment - the use of mathematical algo. to transform data into a form that is not readily intelligible. The transformation and subsequent recovery of the data depend on an algo. and two or more encryption keys.

③ Digital Signature - cryptographic transformation of a data unit that allows a recipient of the data unit to prove the source and integrity of the data unit and protect against forgery.

④ Access Control - a variety of mechanisms that enforce access rights to resources.

⑤ Data Integrity - a variety of mechanisms used to assure the integrity of a data unit or stream of data units.

⑥ Authenticate Exchange - a mechanism intended to ensure the identity of an entity by means of information exchange.

⑦ Traffic Padding - the insert of bits into gaps in a data stream to frustrate traffic analysis attempts.

① Routing Control - enables selects of particular physically secure routes for certain data & allows routing changes, especially when a breach of security is suspected.

② Mediation - the use of a trusted third party to assure certain properties of a data exchange.

③ Pervasive Security Mechanisms - mechanisms that are not specific to any particular OSI security service or protocol layer.

④ Trusted functionality - that which is perceived to be correct with respect to some criteria.

⑤ Security label - The marking bound to a resource that names or designates the security attributes of that resource.

② Event-Detect - detects of security-relevant events.

③ Security Audit Trail - Data collected and potentially used to facilitate a security audit, which is an independent review and examination of system records and activities.

④ Security Recovery - deals with requests from mechanisms, such as event handling and management functions and later recovery actions.

$$P(\text{Secure build}) = P(E)$$

$$P(\text{Priv}) = P$$

$$P(\text{Conf})$$

Q Explain Network Security Model?

Ans

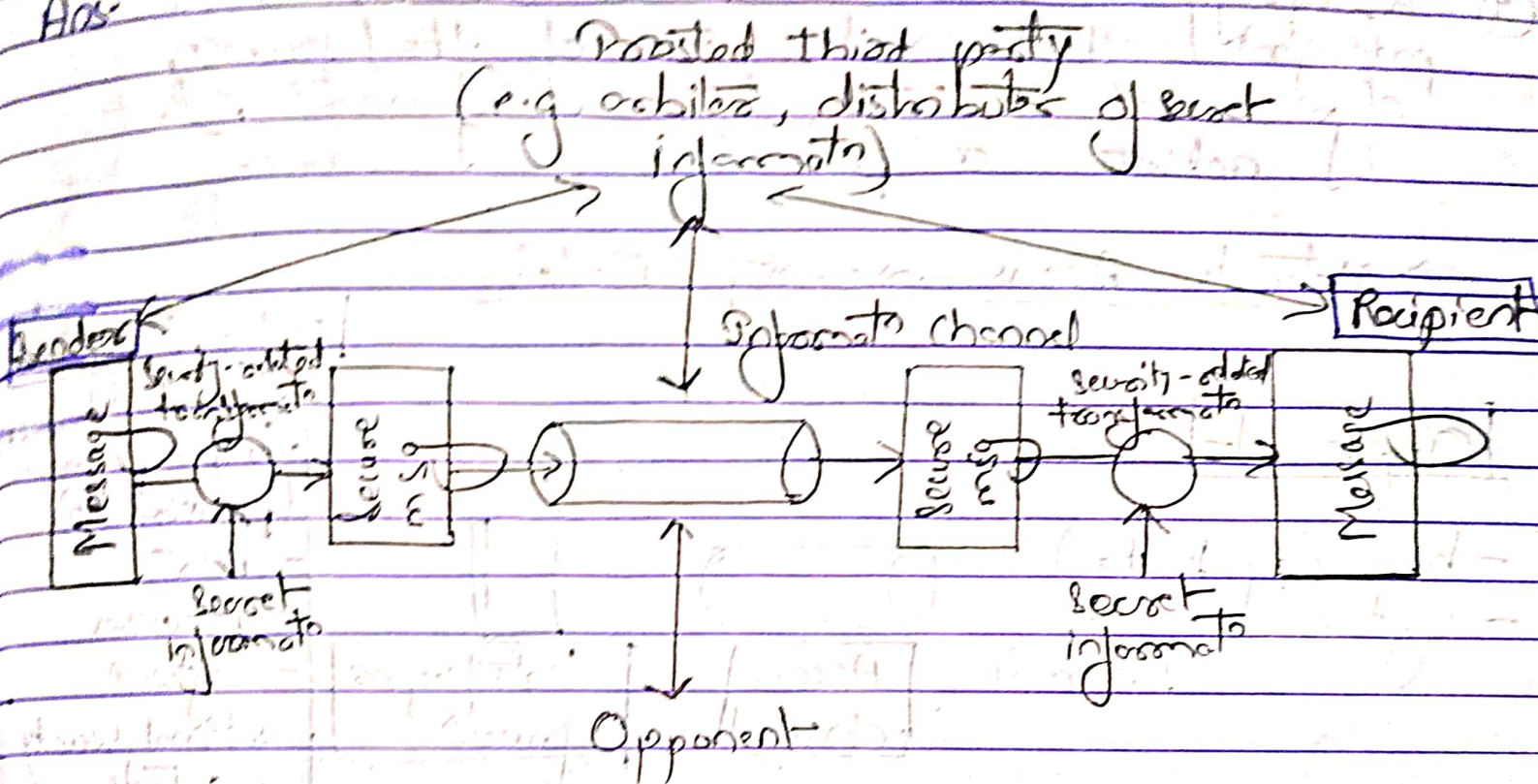


Fig:- Model for Network Security

→ This general model shows that there are four basic tasks in designing a particular security service:-

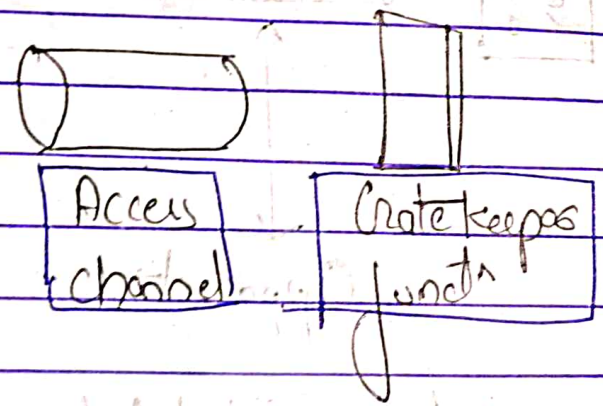
- ① Design an algorithm for performing the security-related transformations.
- ② Generate the secret information to be used with the algorithm.

- ③ Develop methods for the distribution and sharing of the secret information.
- ④ Specify a protocol to be used by two principals that make use of the security algo. and the secret information to achieve a particular service.

→ Network Access Security Model

Opponent:-

- human (eg, hacker)
- software (eg, virus, worms)



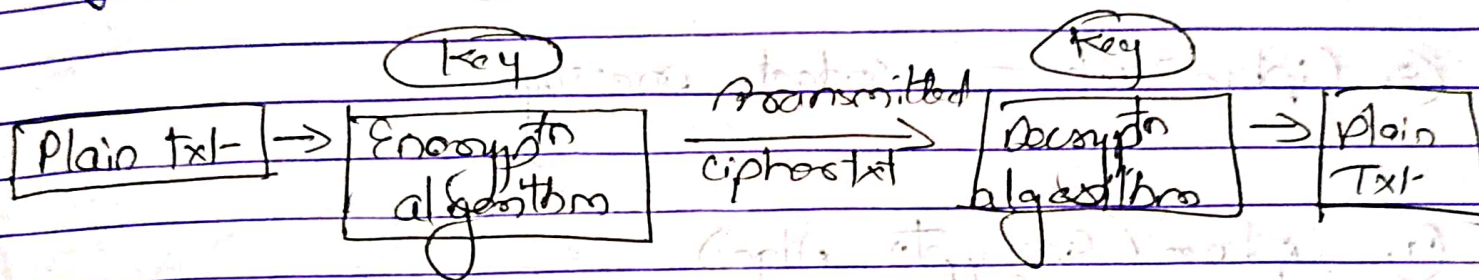
Information system

- Computing resources (processor, I/O)
- Data
- Processes
- Software
- Internal security controls

~~Solve~~ 1) $PC(\text{decrypt}) = PC(E)$
 $PC(E(M)) = PC$

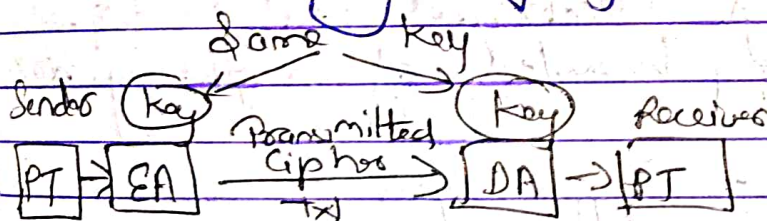
Cryptography :-

The art or science encompassing the principles and methods of transforming an intelligible msg into one that is unintelligible and then retransforming that msg back to its original form.

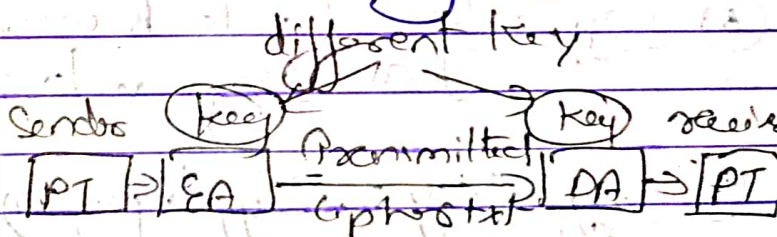


Types

① Symmetric Cryptography (Private Key Cryptography)

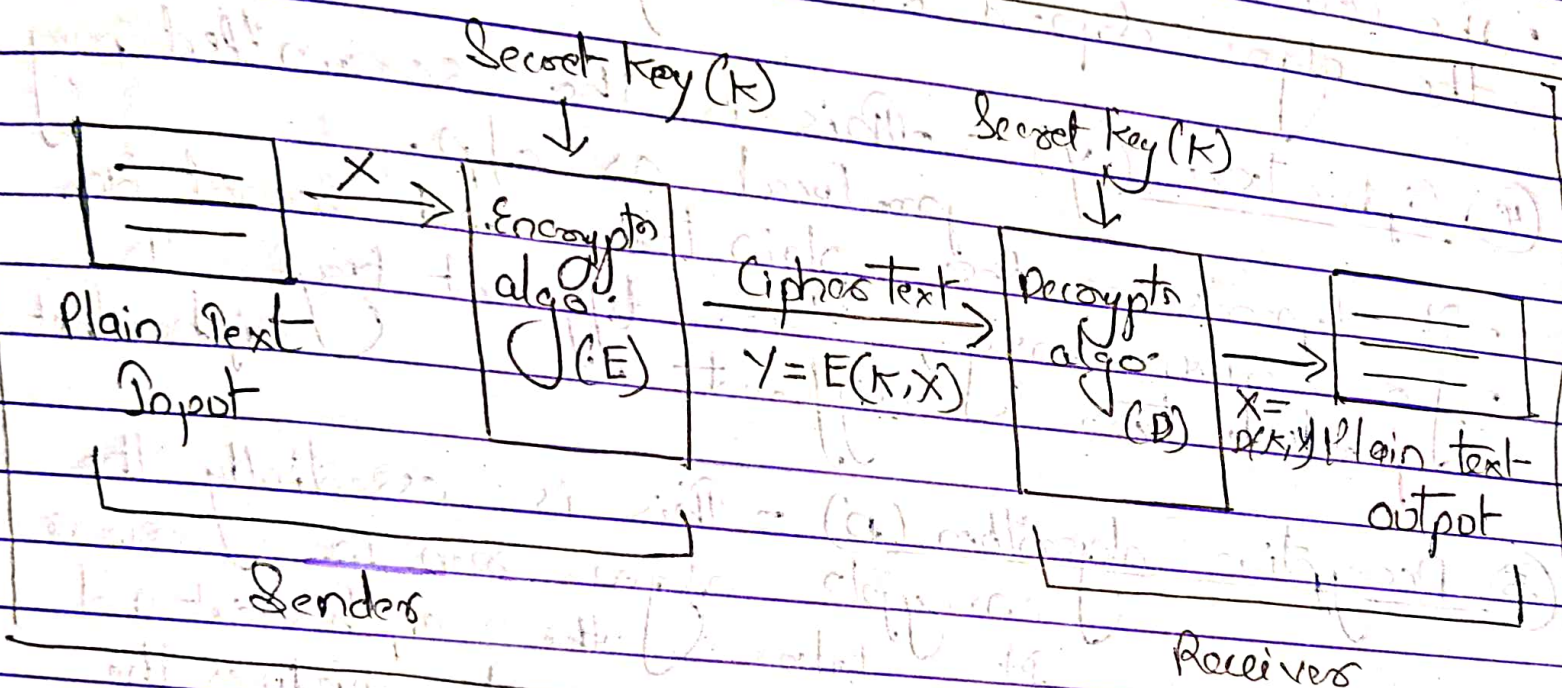


② Asymmetric Cryptography (Public Key Cryptography)



Define Symmetric cipher model?

A symmetric cipher model is composed of five essential parts :-



- ① Plain text (X) - This is the original intelligible msg/data that is fed into the algo. as i/p.
- ② Encrypts algorithm (E) - it performs various substitutions and transformations on the plaintext.
- ③ Secret key (K) - The secret key is also i/p to the encrypts algo.

- The key is a value independent of the plain text and of the algorithm.
- The algorithm will produce a different output depending on the specific key being used at the time.
- The exact substitutions and transformations performed by the algo. depend on the key.

④ Cipher Text (y) - This is the scrambled msg produced as output.

- It depends on the plain text and secret key.
- For a given msg, two different keys will produce two different cipher texts.

⑤ Decryption algorithm (D) - This is essentially the encryption algo run in reverse. It takes the ciphertext and the secret key and produces the original plain text.

Requirement for Encryption :-

- ① Encryption algo. - there is need for a very strong EA.
- ② Secure way to share secret key - there must be some and robust way to share the secret key b/w sender & receiver.

=> Classical Encryption Techniques

① Substitution

- Caesar Cipher
- Monoalphabetic Substitution
- Playfair Cipher ✓
- Hill Cipher ✓
- Polyalphabetic Ciphers
- One-Time Pad

② Transposition

- Rail Fence ✓
- Row Column Transposition

Playfair Cipher

- Aka Playfair Square or Wheatstone-Playfair Cipher
- Manual symmetric encryption technique
- Multiple letter encryption cipher
- Diagrams
- 5x5 matrix constructed using a keyword

Q: Monarchy

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

→ Rules for encrypting using Playfair Cipher

- ① Digrams
- ② Repeating letters - filler letter
- ③ Same column | ↓ | wrap around
- ④ Same row | → | wrap around
- ⑤ Rectangle | ↔ | swap

Example

Plaintext: attack

Digrams: at ta ck

Plaintext: reso academy

Digrams: re so ac ad em yx

Plaintext: balloon

Digrams: ba ll oo n

Digrams: ba lx lo on

→ repeating letter

Q1. Encrypt the following using play fair cipher using the keyword MONARCHY - "SAVARAJ IS MY BIRTH RIGHT"
Use X as blank space

Ans SW AR AJ IS MY BI R

SW	AR	AJ	IS	MY	BI	RT	HR	IG	HT
OX	RM	BS	SX	NC	IS	DZ	DO	KI	DP

following table

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Diagram : OXRMBSXNCISDZDO KIDP

Rail Fence :-

* The plaintext is written down as a sequence of diagonals and then read off as a sequence of rows

Example Encipher the msg "neso academy is the best" with a rail fence of depth 2

Soln

Plaintext - neso academy is the best
Depth - 2

n		s		a		a		e		y
	e		o		c		d		m	

i	s		b		b		s	
		t		e		e		t

Ciphertext: N S A E Y S H B S E O C D M T T E E T

Q2 Convert the Given Text - "CRYPTOGRAPHY" into cipher text using Rail fence Technique

PT: CRYPTOGRAPHY
Depth: 2

C		Y		T		G		A		H	
	R		P		O		R		P		Y

Ciphertext - CYTG AHRPORY

Steganography

Q Explain how steganography can be used to enhance security in communication?

⇒ Steganography :-

- Conceal the existence of msg
- It is like a secret code for hiding msg in plain sight.
- Not an encrypt scheme
- Cryptography renders the msg unintelligible to outsiders by various transformations of the text.
- Ex:- Simply encrypt - correct reading exactly twice
- Examples

① Character marking :- selected letters of printed or typewritten text are over written in pencil.

② Invisible ink :- a no. of substances can be used for writing but leave no visible trace unless the trace until heat or some chemical is applied to the paper.

③ Pin punctures :- Small pin punctures on selected letters are ordinarily not visible unless the paper is held up in front of a light.

④ Typewriter corrector ribbon :- Used b/w lines type with a black ribbon, the results of typing with the corrector tape are visible only under a strong light.

Ans: Steganography enhances security in communication by concealing the existence of the msg itself.
Here's how it works

① Hidden msg : Instead of encrypting data, steganography hides it within ordinary files like image, audio & video.

② Layered security : It can be combined with encryption so that even if the hidden msg is discovered, it remains protected by encryption.

③ Avoids Detection :- Since the msg is hidden within regular files, it reduces the chance of being intercepted by eavesdroppers and malicious parties.

④ Non-disruptive :- The carrier file (image, audio etc) remains functional and appears normal, preventing suspicion.

⑤ Secure Transmission :- It allows secure and covert communication.

Limitations of Perfect Secrecy

Perfect Secrecy - is the notion that, given an encrypted msg (or ciphertext) from a perfectly secure encryption system (or cipher), absolutely nothing will be revealed about the unencrypted msg (or plaintext) by the ciphertext.

→ limitations of perfect secrecy in cryptography :-

① Key Management :- requires a one-time pad as long as the msg, making key generation, storage and management difficult.

② Key Distribution :- securely sharing the key b/w sender & receiver is a challenge, as it must be done over a secure channel.

③ Key Reuse :- Keys can only be used once. Reusing a key compromises the security of the encryption.

④ Practicality :- impractical for large-scale as generating & securely distributing long keys for each msg is not feasible.

⑤ Storage space :- the requirement for a key as large as the msg creates significant overhead in terms of storage space for both the key and the encrypted msg.